

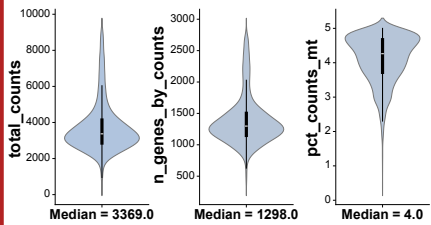
A



.h5 files

Easy ambient RNA removal
(*CellBender*)

Semi-automated quality control
(*Scanpy*, *Seurat*, *scDbtFinder*, *Scrublet*, *DoubletDetection*)



total_counts
Median = 3369.0

n_genes_by_counts
Median = 1298.0

pct_counts_mt
Median = 4.0

Filter based on quantiles
Filter based on absolute cutoffs
Doublet removal
Filter based on mitochondrial content
Evaluate samples correlation

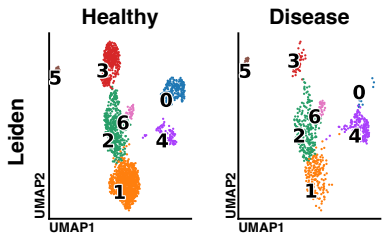
Multi-platform Support
(*Zellkonverter*)

SeuratObject SingleCell Experiment

AnnData

B

Batch integration and clustering



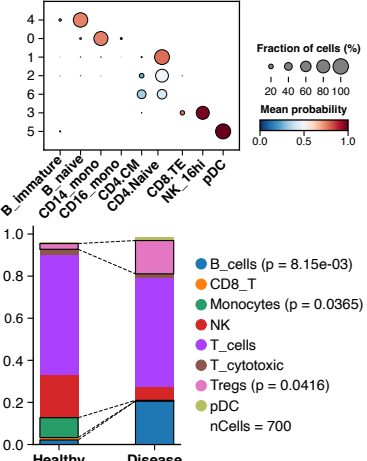
Healthy Disease

Leiden

UMAP1 UMAP2

Harmony, *Scanorama*, *scVI*, *CCA*, *BBKNN*

Identification and evaluation of cell population
(*Celltypist*, *Scanpro*)



Fraction of cells (%)

Mean probability

Healthy Disease

Cell populations: B_immature, B_naive, CD4_T_mono, CD4_CM, CD4_naive, CD8_TE, NK_16n, pDC

Cell population proportions (Healthy vs Disease):

- B_cells (p = 8.15e-03)
- CD8_T
- Monocytes (p = 0.0365)
- NK
- T_cells
- T_cytotoxic
- Tregs (p = 0.0416)
- pDC

nCells = 700

C

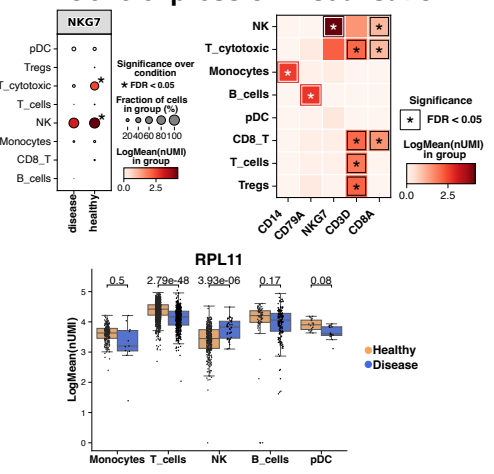
Pseudo-bulk and single-cell differential expression analysis

Pseudobulk
DESeq2
EdgeR
t-test

Consensus

Single-Cell
Wilcoxon
t-test
MAST
LogisticReg

Gene expression visualisation



NKG7

RPL11

LogMean(nUMI)

Healthy Disease

Cell populations: Monocytes, T_cells, NK, B_cells, pDC

Top 10 GO Terms in each Condition

Healthy Disease

NKG7

p = 1.11e-21

LogMean(nUMI)

Healthy Disease